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2023-2024

# Алгоритм ба Өгөгдлийн БҮТЭЦ - ЛАБ

io × UFE



```
function linearSearch(arr, searchValue) {  
  for(let i = 0; i < arr.length; i++) {  
    if(arr[i] === searchValue) {  
      return i;  
    }  
  }  
  
  return -1;  
}  
  
let arr = [64, 25, 12, 22, 11];  
console.log(linearSearch(arr, 22))
```

```
class Student {  
  constructor(id, name) {  
    this.id = id;  
    this.name = name;  
  }  
}  
  
let arr = [];  
arr.push(new Student("1", "Bataa"));  
arr.push(new Student("2", "Boldoo"));  
arr.push(new Student("3", "Bayraa"));  
arr.push(new Student("4", "Sainaa"));  
  
function linearSearchById(arr, searchValue) {  
  for(let i = 0; i < arr.length; i++) {  
    if(arr[i].id === searchValue) {  
      return i;  
    }  
  }  
  
  return -1;  
}  
  
const studentIndex = linearSearchById(arr, "3");  
console.log(arr[studentIndex]);
```

```
function binarySearch(arr, l, r, x){
  if (r >= l) {
    let mid = l + Math.floor((r - l) / 2);

    if (arr[mid] == x)
      return mid;

    if (arr[mid] > x)
      return binarySearch(arr, l, mid - 1, x);

    return binarySearch(arr, mid + 1, r, x);
  }

  return -1;
}

let arr = [ 2, 3, 4, 10, 40 ];
let x = 10;
let n = arr.length
let result = binarySearch(arr, 0, n - 1, x);
(result == -1) ? console.log( "Element is not present in array")
               : console.log("Element is present at index " +result);
```

<https://www.geeksforgeeks.org/breadth-first-search-or-bfs-for-a-graph/>

**DFS-ийг хөгжүүлж ирэх**